

Chikungunya transmission with explicit demographic accounting, Ceará, Brazil

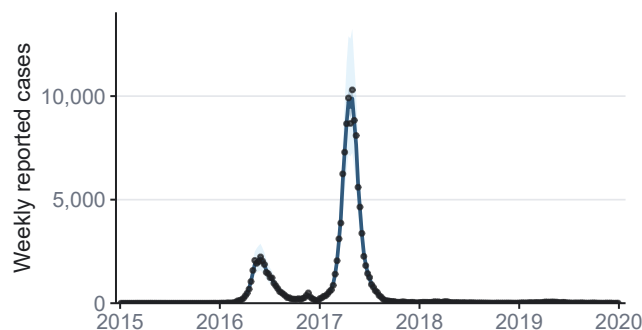
Bayesian renewal model v2.2, weekly observations from 2015 to 2019

A

Reported cases and posterior prediction

Shading: 95% predictive interval; dark band: 50% CrI

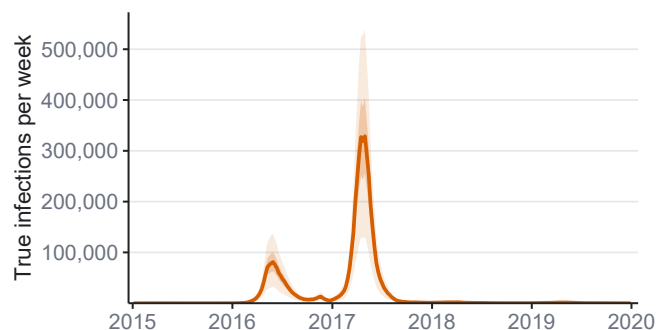
— Model expectation • Observed cases



B

Latent infection incidence

Median, 50% and 95% credible intervals

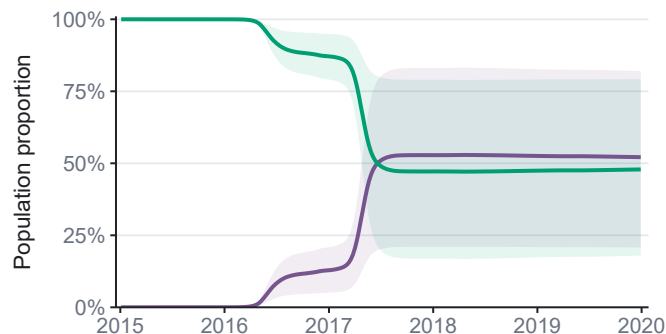


C

Population susceptibility and immunity

S, U tracked explicitly from births/deaths/reconciliation

— Infection-derived immune — Susceptible

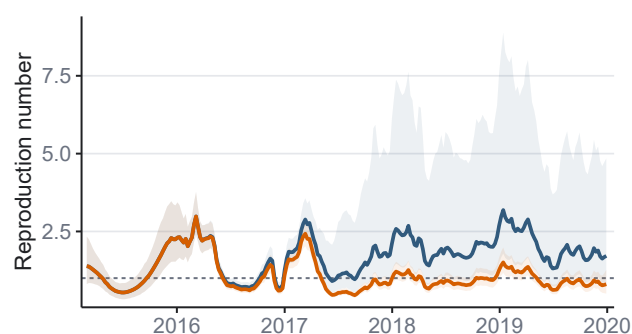


D

Time-varying reproduction numbers

$$R_{\text{eff}}(t) = R_0(t) \times S(t) / (S(t) + U(t))$$

— $R_0(t)$ — $R_{\text{eff}}(t)$

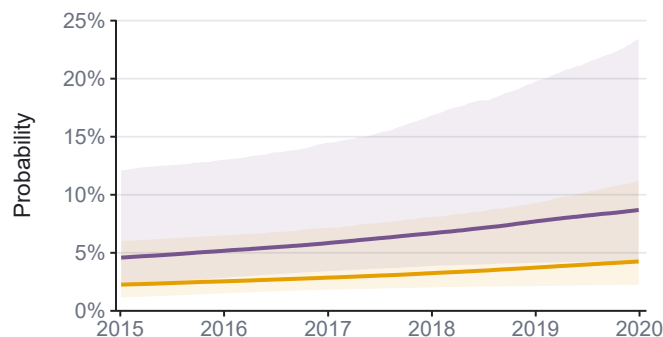


E

Case ascertainment

Estimated symptomatic proportion: 0.50 [0.37, 0.61]

— Infection-to-report — Symptomatic-case reporting



F

Cumulative infection and serological anchors

Shaded fields: Juazeiro do Norte (orange) and Quixadá (teal)

— Ceara — Site model — Site observed

